Progressive exercises

E1 · Direct application. In a coordinate system of space, consider $A(1;-1;2)$ and $B(3;0;1)$. Give a parametric representation of the line (AB).

E2 · Application and justification. In an orthonormal coordinate system, determine a Cartesian equation of the plane passing through $A(1;2;3)$ and normal vector $\vec{n} = (1;0;-2)$.

E3 · Method to organise. In a coordinate system of space, the line d is given by $(x;y;z) = (t;1+t;2-t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x + 2y + z = 8$. Determine their intersection.

E4 · Choice of strategy. In a coordinate system of space, determine the intersection of the planes with equations $x + y + z = 1$ and $x - y = 0$. Give a parametric representation if the intersection is a line.

E5 · Synthesis or proof. In a coordinate system of space, study the relative position of the planes with equations 2 $x - y + z = 3$ and $4x - 2y + 2z = 5$. Justify.